Exercice 1.

Questions 1

1 SELECT STATEMENT.

Write a SQL query to Retrieve All Columns From Employees table.

| ID | First_Name | Last_Name | department | salary | hire_date | City |
|---|---|---|---|---|---|---|
| 1 | John | Doe | I I | 55,000 | 2018-06-15 | New York |
| 2 | Jane | Smith | H R | 48000 | 2019-07-20 | Chicago |
| 3 | Mike | Johnson | Finance | 60000 | 2017-09-30 | Los-Angeles |
| 4 | Sarah | Brown | I T | 53 000 | 2021-03-25 | New York |
| 5 | David | white | Marketing | 52 000 | 2016-04-10 | San Francisco |
| 6 | Emily | Davis | I T | 62 000 | 2015-02-14 | Chicago |
| 7 | Robert | wilson | Finance | 59000 | 2019-10-01 | Houston |
| 8 | Jessica | Moore | H R | 51000 | 2018-05-22 | Los-Angels |
| 9 | Daniel | Clark | Marketing | 53 000 | 2022-06-01 | Chicago |
| 10 | Laura | Hall | I T | 50000 | 2020-08-10 | San Francisco |

Syntax

```
select *
FROM Employees.
```

2. SELECT DISTINCT statement.

Write a SQL query to find All unique departments in the Employees Table.

```
SELECT department
FROM Employees
```

| Department |
|---|
| I T |
| Finance |
| HR |
| Marketing. |

3 ORDER BY Statement.

Write a SQL query to Retrieve All Employees First and Last names, ordered By Salary in descending order.

Syntax

```
SELECT  First_Name,
        LAST_Name,
        Salary
From Employees
Where Condition
ORDER BY (DESC);
```

| First_Name | LAST_Name | Salary |
|---|---|---|
| Emily | Davis | 62000 |
| MIKE | Johnson | 60000 |
| Robert | wilson | 59000 |
| John | Doe | 55000 |
| Sarah | Brown | 53000 |
| David | white | 52000 |
| Jessica | Moore | 51000 |
| Laura | Hall | 50000 |
| Jane | Smith | 48000 |

4 limit statement
write a SQL Query to Retrieve the top 5 highest Paid Employees.

```
SELECT *
FROM Employees
LIMIT 5;
```

| ID | First_Name | LAST_Name | Department | Salary | hiredate | City |
|---|---|---|---|---|---|---|
| 6 | Emily | Davis | IT | 62000 | 2015-02-14 | Chicago |
| 3 | Mike | Johnson | Finance | 60000 | 2017-09-30 | Los-Angeles |
| 7 | Robert | Wilson | Finance | 59000 | 2019-10-01 | Houston |
| 1 | John | Doe | IT | 55000 | 2018-06-15 | New-York |
| 4 | Sarah | Brown | IT | 53000 | 2021-03-25 | New-York |

5. Where statement.
Write a SQL query to Find employees who work in the IT department.

```
SELECT *
From employees
where department = 'IT';
```

| ID | First_Name | LAST_Name | department | Salary | hire_date | City |
|---|---|---|---|---|---|---|
| 1 | John | Doe | IT | 55000 | 2018-06-15 | New-York |
| 4 | Sarah | Brown | IT | 53000 | 2021-03-25 | New-York |
| 6 | Emily | Davis | IT | 62000 | 2015-02-14 | Chicago |
| 10 | Laura | Hall | IT | 50000 | 2020-08-10 | San-Francisco |

6 AND STATEMENT

write a SQL Query to Find employees who work in the Finance department AND have a salary greater than 58,000.

```
SELECT  First_Name,
        LAST_Name,
        ~~Department~~ ~~Salary~~ Department
From EMPLOYEES
where Salary > 58,000 ;
```

| First_Name | LAST_Name | Department | Salary |
|---|---|---|---|
| MIKE | JohnSon | Finance | 60000 |
| Robert | Wilson | FINANCE | 59000 |

7 OR Statement

Write a SQL query to Find Employees Who work in the HR department OR the Marking department.

```
SELECT *
FROM EMPLOYEES
where department = 'HR' OR
department 'Marketing';
```

| | ID | First_Name | LAST_Name | Department | Salary | Hire_date | City |
|---|---|---|---|---|---|---|---|
| | 2 | Jone | Smith | HR | 48000 | 2019.07.20 | Chicago |
| | 8 | Jessica | Moore | HR | 51000 | 2018.05.22 | Las Angele |
| 5 | David | white | Marketing | Marketing | 52000 | 2016_04.10 | San Franasco |
| | 9 | Daniel | Clark | Marketing | 58000 | 2022.06.1 | Chicago |

8. Not statement
write a SQL Query to find employees who do not work in the IT department.

```
SELECT *
FROM Employees
where Not department = 'IT';
```

| ID | First_Name | LAST_Name | Department | Salary | hire_date | City |
|---|---|---|---|---|---|---|
| 2 | Jane | Smith | HR | 48000 | 2019-07.20 | chicago |
| 3 | MIKE | Johnson | Finance | 60000 | 2017.09.20 | Los Angeles |
| 5 | David | ~~Wilhite~~ | Marketing | 52000 | 2016.04.10 | San Fransuco |
| 7 | Robert | Wilson | Finance | 59000 | 2019.10.01 | Houston |
| 8 | Jessica | Moore | HR | 51000 | 2018.06.01 | Los Angels |
| 9 | Donid | Clark | Marketing | 53000 | 2022.06.01 | Chicago |

9 IN Statement
Write a SQL query to find employees who are in the H.R. I.T. or Finance departments.

```
SELECT *
from employees
where department IN ('HR','IT','FINANCE');
```

| ID | First_Name | LAST_Name | department | Salary | hide_date | City |
|---|---|---|---|---|---|---|
| 2 | Jane | Smith | HR | 48000 | 2019.07.20 | chicago |
| 8 | Jessica | MOORE | HR | 51000 | 2018.05.02 | Los. Angeles |
| 1 | John | Doe | IT | 55000 | 2018.06.15 | New Jork |
| 4 | Sarah | Brown | IT | 53000 | 2021.03.25 | New Jork |
| 6 | Emily | Davis | IT | 62000 | 2015.03.14 | Chugoco |
| 10 | LAURA | Hall | IT | 50000 | 2020.08.10 | San Francisco |
| 3 | MIKE | Johnson | Finance | 60000 | 2017.09.30 | LAS Angeles |
| 7 | Robert | Wilson | Finance | 59000 | 2019.10.01 | Houston |

10 Combining Conditions.

Write a SQL query to find employees who are in the IT department, have a Salary greater than 50.000 and are located in new york.

```
SELECT *
From Employees
where Department = IT
AND Salary > 50.000
AND Located = NEW YORK ;
```

| ID | First_name | LAST_name | department | Salary | hire_date | City |
|---|---|---|---|---|---|---|
| 1 | John | Doe | IT | 55000 | 2018-06-15 | New York |
| 4 | Sarah | Brown | IT | 53000 | 2021-03-25 | New York |
| 6 | Emily | Davis | IT | 62000 | 2015-02-14 | Chicago |

11

```
SELECT First Names,
       LAST Names,
From employees
where Department ('Finace' OR 'Marketing')
AND Salary ≥ 52000
ORDER BY (DESC);
```

(11)

~~Total Names~~

| First_Names | LAST_Name | department | salary |
|---|---|---|---|
| MIKE | John son | Finance | 60000 |
| Robert | Wilson | Finance | 59000 |
| Donel | Clark | Marketing | 53000 |
| | | | |

12

SELECT ~~Cities~~ Distinct (Cities)
FROM Employees
~~where Department (excluding those in IT and HR)~~
~~IN~~
Where departme NOT IN ('IT', 'HR');

| Cities |
|---|
| Los Angels |
| Houston |
| San Francisco |
| Chicago |